

Shared Memory and Memory-Mapped File

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Definition

Inter-Process Communication (IPC) is a mechanism that allows two or more processes to communicate and exchange data with each other. It helps processes coordinate and share information while running independently.

Shared Memory and Memory-Mapped File

Shared Memory allows multiple processes to access the same area of memory. It is a fast IPC technique because processes can directly read and write data in the shared memory.

Memory-Mapped File maps a file into the memory of a process. Multiple processes can access the same mapped file, allowing them to share and modify data efficiently.

Example

Imagine a **banking application** where one process updates an account balance and another process displays it. Both processes can use shared memory to access the latest balance quickly.

For a **memory-mapped file**, a text file containing customer information can be mapped into memory so that multiple processes can read or update the information without repeatedly reading the entire file.